

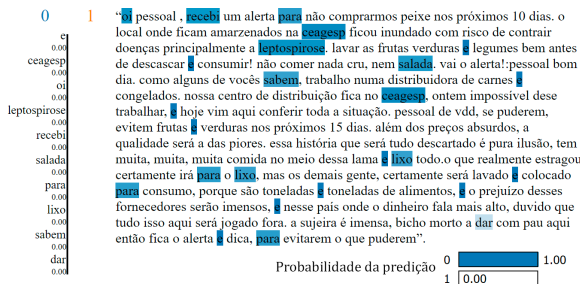
1 Analysis of the Explanations Generated by LIME and by IG

1.1 Explanations Generated by LIME

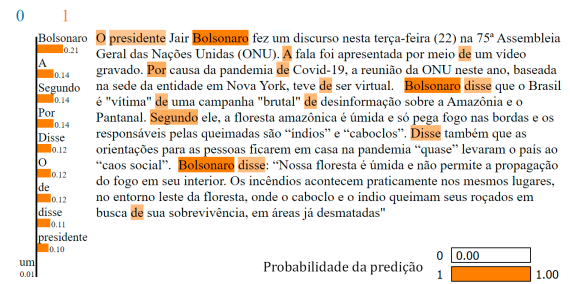
In this section, the explanations generated by LIME were analyzed. Four explanations of the samples classified by each model were highlighted and discussed in the text, representing each of the selection criteria: a fake news sample correctly classified as fake; a forgery incorrectly classified as real; a real news correctly classified as real; and finally, a real news classified as fake. The explanations generated by IG were analyzed following the same strategy and can be found in Section 1.2.

1.1.1 Model 1: FakeRecogna including stop words

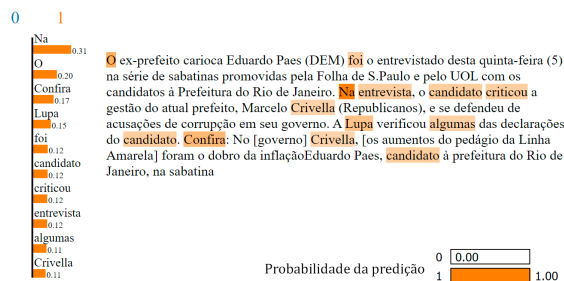
Figure 1 – Explanations generated by LIME for Model 1



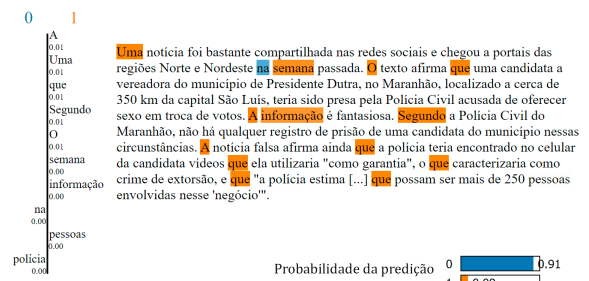
(a) Fake classified as false



(b) Fake classified as real



(c) Real classified as real



(d) Real classified as fake

1.1.1.1 Fake news classified as fake

In Figure 1a, there is a LIME explanation for a fake news with stop words, correctly classified by Model 1. Based on it, the following points can be observed:

- Reference to health (saúde): the words *leptospirose*, *salada* (salad), and *lixo* (trash), which were highlighted with negative scores, are words that address health (saúde) and sanitary conditions. The use of these words in a negative context may increase suspicion that the news is fake (falsa).
- Personal opinions (opiniões pessoais): the words *recebi* (I received) and *oi* (hi) were highlighted with a negative score; their use may indicate a more personal narrative, and these opinions can influence the interpretation of the news' truthfulness.
- *e* (and): it is a very common conjunction in Portuguese, considered a stop word (palavra de parada), generally not carrying specific semantic meaning, but it presented a negative score.
- *Para* (for/to): it is also a stop word (palavra de parada) and a multifunctional preposition in Portuguese and also received a negative score.
- *ceagesp*: The negative score assigned to this word may indicate that the mention of the organization Ceagesp could be associated with negative events.

1.1.1.2 Fake news classified as real

Analyzing the LIME explanation in Figure 1b for a fake news incorrectly classified as real, the following can be observed:

- *A, O, de*: in this example, it is noticeable that common words in the Portuguese language were highlighted with positive weights. The positive weight of these words alone does not provide much information about the classification, but it may lead us to consider that real news are better written, and that in many fake news, these basic words considered stop words (palavras de parada) are scarce in the text structure. The presence of these words with positive weights can also be explained by the fact that they are considered “neutral” in terms of informational content.

- **Bolsonaro** and **presidente** (president): these words had positive weights, which may indicate a political inclination.
- **um** (a/an): it received a negative score, but very small compared to the other words.

1.1.1.3 Real news classified as real

Based on the LIME explanation in Figure 1c, referring to a real news sample classified as real:

- **Na**: it received a high positive score. In this context, it may indicate the beginning of a sentence or introduce a temporal location, contributing to the structuring and cohesion of the news.
- **O** and **foi**: again, common words in the Portuguese language received positive scores.
- **Lupa**: it received a positive score. This may indicate fact-checking performed by a reliable source, adding credibility to the news.
- **Candidato** (candidate): it received a positive score. Its presence indicates the identity of the interviewee, providing essential information for understanding the news.
- **Crivella**: the positive score may indicate a political inclination.

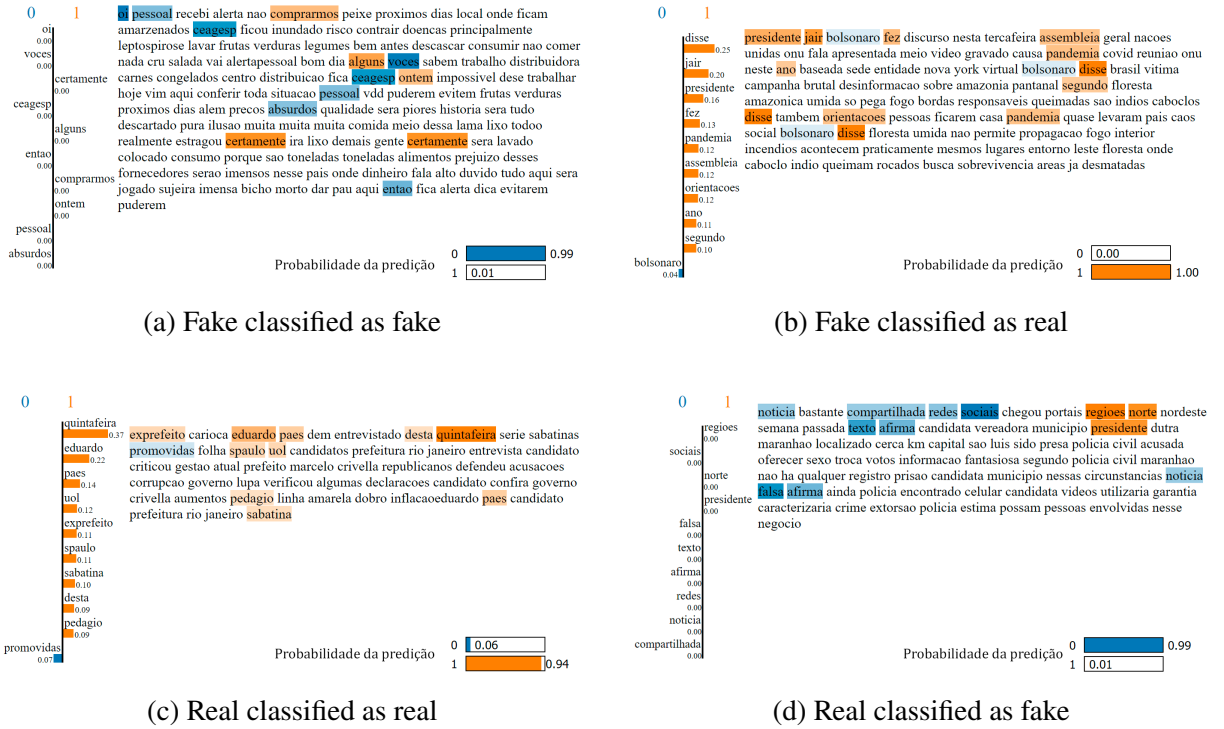
1.1.1.4 Real news classified as fake

Finally, according to Figure 1d, which presents the LIME explanation for a real news classified as fake:

- **A**, **O**: two articles received a positive score.
- **na**: this word is highlighted once again, but now it received a negative score.
- **Segundo** (according to): it received a positive score. This word generally introduces information from reliable sources, such as statements from police authorities, increasing the credibility of the news.
- **Semana** (week): this word received a positive score, possibly because it indicates a time period in which the news was shared, providing temporal information that makes the news more plausible.

1.1.2 Model 2: FakeRecogna without stop words

Figure 2 – Explanations generated by LIME for Model 2



1.1.2.1 Fake news classified as fake

The explanation generated by LIME in Figure 2a refers to the same fake news sample previously analyzed as classified as fake, but now without the use of stop words and classified by Model 2, leading to the following observations:

- **ceagesp**: once again, the name of this organization received a negative score, which may indicate that this organization is a frequent target of fake news.
- **oi**: a common greeting in Portuguese received a strong negative score. In this context, this may be due to the fact that news produced by a responsible channel does not use informal language.
- **pessoal** (guys/people) and **voces** (you all): these words, also used in informal communication, received negative scores.
- **certamente** (certainly): it received a positive score. Although **certamente** is a positive word, not much information can be inferred from its isolated usage.

1.1.2.2 Fake news classified as real

In Figure 2b, the LIME analysis of a fake news that was incorrectly classified as real is shown, where the following can be observed:

- **presidente**: reference to the president of the republic received a positive weight.
- **jair** and **bolsonaro**: it is curious to note that the first name of the former president **jair** received a positive weight for the news, favoring the prediction as real, while the surname **bolsonaro** received a negative weight. This might indicate that when the former president is the target of fake news, the outlets that create them tend to use only his surname.
- **pandemia**: reference to the Covid pandemic received a positive weight in the news.
- **ano**: once again, a temporal reference received a positive weight for the prediction.

1.1.2.3 Real news classified as real

In Figure 2c, the LIME analysis of a real news classified as real is shown, where the following can be observed:

- **exprefeito**: reference to a politician received a positive weight for the news.
- **eduardo** and **paes**: reference to the name of the former mayor of Rio de Janeiro obtained a positive weight, which may indicate a certain political inclination.
- **quinta-feira** (Thursday): a temporal reference that received a positive weight for the prediction.
- **uol**: a news outlet received a positive score.
- **promovidas** (promoted): it received a very small negative weight, but no significant conclusions can be drawn from this word.

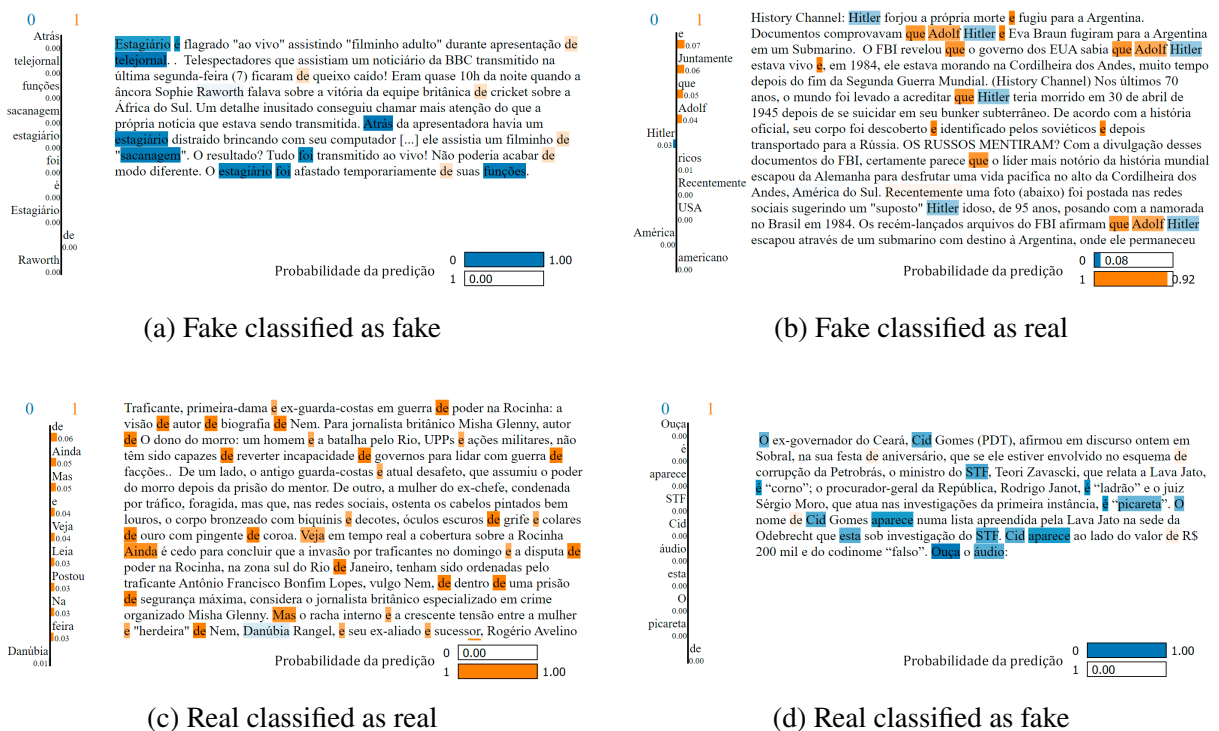
1.1.2.4 Real news classified as fake

Analyzing Figure 2d, which contains the LIME analysis of a real news that was classified as fake, the following observations can be made:

- **compartilhada** (shared), **redes** (networks), **sociais** (social): references to the sharing of news on social networks received negative weights, which may be because fake news is often spread through these networks.
- **notícia** (news): the word **notícia** alone received a negative weight. This might suggest that fake news often explicitly uses this word as an appeal to grab attention. However, in this case, the news is real and refers to a fake news.
- **falsa** (false): this word had a high negative weight for the classification, as it literally refers to fake news. Nevertheless, the news is real and the topic addressed was a news item that had been confirmed as false.
- **presidente** (president): a reference to a political figure once again received a positive weight.
- **regiões** (regions) and **norte** (north): mentioning the northern regions received a positive weight, which may indicate that locality information can increase the credibility of the news. However, in contrast, the word **nordeste** (northeast) did not show any prominence.

1.1.3 Model 3: Fake.Br including stop words

Figure 3 – Explanations generated by LIME for Model 3



1.1.3.1 Fake news classified as fake

Analyzing now the explanations of the samples classified by Model 3, in Figure 3a is the explanation generated by LIME of a fake news sample classified as fake from the Fake.Br Corpus *dataset* with *stop words*, which led to the following observations:

- **estagiario**: the reference to this entity received a negative score, but it was not possible to derive many insights as to why.
- **telejornal**: it received a negative score, which may indicate that the creators of fake news included the word **telejornal** as part of a strategy to mislead readers, making them think the information comes from a trustworthy source.
- **é**: a very common word in Portuguese considered a *stop word* that can be used in various contexts received a negative score. In some cases, it may be used in statements that are factually true, while in others it may be used to convey false claims. The presence of the word **é** in fake news may be associated with misleading assertions attempting to appear truthful.
- **de**: again a *stop word* received a positive score.

1.1.3.2 Fake news classified as real

The explanation generated by LIME in Figure 3b about a fake news sample classified as real from the Fake.Br Corpus *dataset* with *stop words* led to the following observations:

- **Adolf** and **Hitler**: the word **Hitler** received a negative score; this word is strongly associated with negative historical events. Therefore, its presence in news can be interpreted as indicative of sensationalist, controversial, or potentially misleading content. On the other hand, the word **Adolf** alone may not have as many negative connotations and can be used in more neutral or even positive contexts, such as referring to people with this name unrelated to the Nazi leader.
- **que** and **e**: both words are common connectors in Portuguese frequently used to link sentences and ideas in a text. Their presence may indicate a more cohesive and well-constructed grammatical structure, often associated with real and informative news.

1.1.3.3 Real news classified as real

In Figure 3c, the LIME explanation of a real news sample classified as real is shown:

- **de, e** and **na**: once again, simple Portuguese words received positive scores, which may reinforce the idea that real news is better written grammatically and makes greater use of these connectors.
- **Veja**: the word **Veja** received a positive score; in this context, it is a conjugation of the verb “to see,” but the positive score is likely due to **Veja** also being interpreted as a reference to the Brazilian magazine of the same name, known for its investigative and reliable journalism.
- **Danúbia**: a proper name received a negative score.

1.1.3.4 Real news classified as fake

Analyzing the real news sample classified as fake by LIME in Figure 3d, the following was observed:

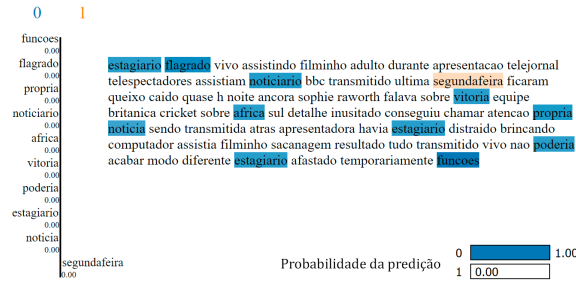
- **o, e**: in this case, the word **e** received a negative score, unlike the previous news. The word **o** also received a negative score; it is an article in Portuguese.
- **de**: as in other news, it also received a positive score.
- **Cid**: reference to a political entity received a negative score.
- **STF**: reference to the Supreme Federal Court, which may indicate it is frequently a target of fake news.
- **picareta**: received a negative weight, which may be because its presence in the news indicates a negative bias or hostility toward the subject, potentially affecting the perception of the credibility of the information.

1.1.4 Model 4: Fake.Br without stop words

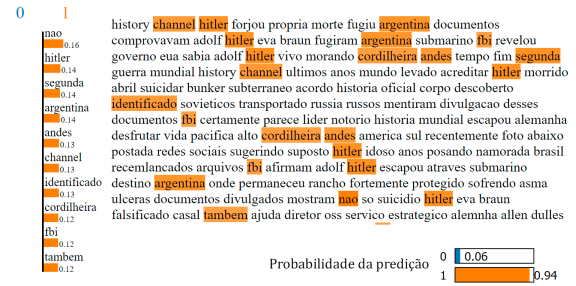
1.1.4.1 Fake news classified as fake

Analyzing the samples classified by Model 4, in Figure 4a is the LIME explanation of a fake news sample correctly classified, where it was possible to observe:

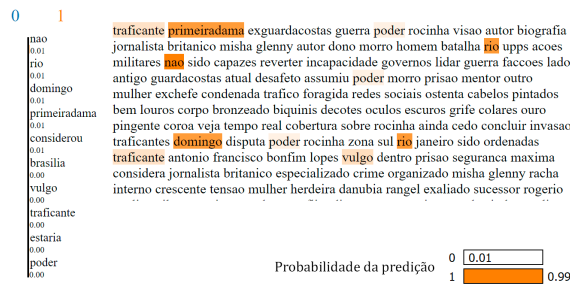
Figure 4 – Explanations generated by LIME for Model 4



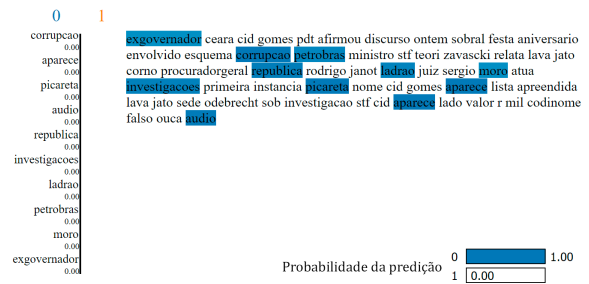
(a) Fake classified as fake



(b) Fake classified as real



(c) Real classified as real



(d) Real classified as fake

- **estagiário** and **telejornal**: these words again received negative scores as in the previous model.
- **noticiario**: it is curious that the word **noticiario**, as well as **telejornal**, received a negative score, since these are outlets where real news is usually published.
- **africa**: reference to a country received a negative score, which may indicate that this country is frequently a target of fake news.
- **segundafeira**: the compound word 'segunda-feira', after removal of *stop words*, was considered a simple word 'segundafeira'; this word received a positive score, possibly because real news tends to be more precise and inform when a fact occurred.

1.1.4.2 Fake news classified as real

Based on Figure 4b, the LIME explanation of a fake news sample classified as real shows:

- **hitler**: in this model the word **Hitler** received a positive score;
- **argentina**, **cordilheira**, and **andes**: reference to a country and the Andes mountain range received positive scores.

- **fbi**: reference to the United States domestic intelligence and security service received a positive score.

1.1.4.3 Real news classified as real

In Figure 4c is the LIME explanation of a real news sample classified as real:

- **primeiradama**: reference to the first lady received a positive score.
- **traficante**: it is common to find news published by reliable sources about trafficking, so perhaps this word has a positive weight because it is a topic often present in real news.
- **não**: a word with a negative connotation, it received a positive score.

1.1.4.4 Real news classified as fake

The LIME explanation of the real news sample classified as fake is in Figure 4c, where it can be noted:

- **exgovernador** and **moro**: reference to political figures received negative weights.
- **corrupção**: corruption received a negative weight, which may indicate that this topic is frequently used to spread fake news about political figures.
- **petrobras**: reference to the company Petrobras received a negative weight, which may indicate that it is a frequent target of fake news.

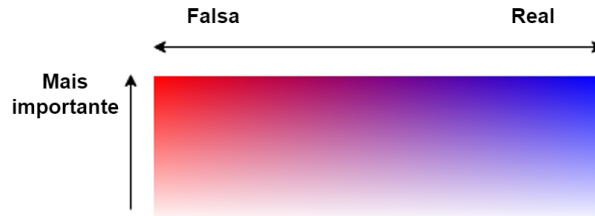
1.2 Explanations generated by IG

This section presents the explanations generated by IG. They use a color scale to highlight the weight of the words, as shown in Figure 5. The redder the word, the more it contributed to the classification as fake news, and the bluer it is, the more it contributed to the news being classified as real, with more saturated colors indicating greater importance of the word's weight for the classification. The application of this color scale to the samples can be seen, for example, in Figure 6.

It is important to emphasize that IG returns a relevance score for each subtoken in the input. However, for better visualization and comparison with LIME, this work considered a score

per *token* (not *subtoken*). To do this, the maximum relevance among all *subtokens* of a *token* was considered to represent the entire *token*.

Figure 5 – Color scale of saliency maps



1.2.1 Modelo 1: FakeRecogna incluindo stop words

Figure 6 – Explanations generated by IG of Model 1

"oi pessoal , recebi um alerta para não comprarmos peixe nos próximos 10 dias , o local onde ficam armazenados na ceagesp ficou inundado com risco de contrair doenças principalmente a leptospirose . lavar as frutas verduras e legumes bem antes de descascar e consumir ! não comer nada cru , nem salada . vai o alerta ! ; pessoal bom dia , como alguns de vocês sabem , trabalho numa distribuidora de carnes e congelados . nossa centro de distribuição fica no ceagesp , ontem impossível dese trabalhar , e hoje vim aqui conferir toda a situação . pessoal de vdd , se puderem , evitem frutas e verduras nos próximos 15 dias . além dos preços absurdos , a qualidade será a das piores . essa história que será tudo descartado é pura ilusão , tem muita , muita , muita comida no meio dessa lama e lixo todo , o que realmente estragou certamente irá para o lixo , mas os demais gente , certamente será lavado e colocado para consumo , porque são toneladas e toneladas de alimentos , e o prejuízo desses fornecedores serão imensos , e nesse país onde o dinheiro fala mais alto , duvido que tudo isso aqui será jogado fora . a sujeira é imensa , bicho morto a dar com pau aqui então fica o alerta e dica , para evitarem o que puderem " .

(a) Fake classified as fake

O presidente Jair Bolsonaro fez um discurso nesta terça - feira (22) na [UNK] Assembleia Geral das Nações Unidas (ONU) . A fala foi apresentada por meio de um vídeo gravado . Por causa da pandemia de Covid - 19 , a reunião da ONU neste ano , baseada na sede da entidade em Nova York , teve de ser virtual . Bolsonaro disse que o Brasil é " vítima " de uma campanha " brutal " de desinformação sobre a Amazônia e o Pantanal . Segundo ele , a floresta amazônica é úmida e só pega fogo nas bordas e os responsáveis pelas queimadas são " índios " e " caboclos " . Disse também que as orientações para as pessoas ficarem em casa na pandemia " quase " levaram o país ao " caos social " . Bolsonaro disse : " Nossa floresta é úmida e não permite a propagação do fogo em seu interior . Os incêndios acontecem praticamente nos mesmos lugares , no entorno leste da floresta , onde o caboclo e o índio queimam seus roçados em busca de sua sobrevivência , em áreas já desmatadas " .

(b) Fake classified as real

O ex - prefeito carioca Eduardo Paes (DEM) foi o entrevistado desta quinta - feira (5) na série de sabatinas promovidas pela Folha de S . Paulo e pelo UOL com os candidatos à Prefeitura do Rio de Janeiro . Na entrevista , o candidato criticou a gestão do atual prefeito , Marcelo Crivella (Republicanos) , e se defendeu de acusações de corrupção em seu governo . A Lupa verificou algumas das declarações do candidato . Confira : No [governo] Crivella , [os aumentos do pedágio da Linha Amarela] foram o dobro da inflação Eduardo Paes , candidato à prefeitura do Rio de Janeiro , na sabatina

(c) Real classified as real

Uma notícia foi bastante compartilhada nas redes sociais e chegou a portais das regiões Norte e Nordeste na semana passada . O texto afirma que uma candidata a vereadora do município de Presidente Dutra , no Maranhão , localizado a cerca de 350 km da capital São Luís , teria sido presa pela Polícia Civil acusada de oferecer sexo em troca de votos . A informação é fantasiosa . Segundo a Polícia Civil do Maranhão , não há qualquer registro de prisão de uma candidata do município nessas circunstâncias . A notícia falsa afirma ainda que a polícia teria encontrado no celular da candidata vídeos que ela utilizaria " como garantia " , o que caracterizaria como crime de extorsão , e que " a polícia estima [...] que possam ser mais de 250 pessoas envolvidas nesse ' negócio ' " .

(d) Real classified as fake

1.2.1.1 Fake news classified as fake

Analyzing the explanations of the samples classified by Model 1, in Figure 6a there is a fake news classified as fake, and it is possible to perceive through the colors and intensities applied by IG:

- ***asburdos*** (absurd): This word obtained a more negative weight; words denoting exaggeration may be common in fake news, as they often try to attract the reader's attention with surprising or incredible claims.
- ***armazenados*** (stored) and ***descascar*** (to peel): these words stood out greatly; however, it is not possible to draw many conclusions from the use of these words.
- ***lavar*** (to wash): this word obtained a positive weight, possibly because it is often used in real news about corruption and money laundering, although in this news it is applied in a different context.
- ***oi*** (hi/hello): it received an important positive weight; it may be correlated with other words or linguistic structures that are more frequent in true news than in fake, but it is not possible to conclude what this correlation is.
- ***prejuízo*** (loss/damage): this word received a positive weight and may be associated with legitimate financial or economic news.
- ***leptospirose*** (leptospirosis): as in the LIME explanations, this word had a negative score.

1.2.1.2 Fake news classified as real

In Figure 6b there is a fake news classified as real, and with the explanation generated by IG it is possible to observe that most words had very similar weights, centered on the color scale and showing a purple color, but the following stood out:

- ***O*** (the [masculine article]): This word is a *stop word* and had a high positive weight. This may reinforce the theory that real news is better written and therefore has more words like this.
- ***Segundo*** (according to): this word also received a higher positive score. The term “segundo” can be used to introduce quotes or information from reliable sources, as used in this news. Thus, this word may be used more often in real news.
- ***Bolsonaro*** (Bolsonaro [proper name]): a political figure had a somewhat higher importance compared to other words, as can be seen by the greater color saturation, having a more positive weight. This characteristic may indicate a political inclination.

1.2.1.3 Real news classified as real

In Figure 6c there is a real news classified as real, and from it the following can be observed:

- ***inflaçãoEduardo*** (inflationEduardo): two words were considered as a single *token* — “inflação” (inflation) and “Eduardo” (Eduardo) — having a more positive weight and greater importance for the sample’s prediction.
- ***UOL*** (UOL [proper name]): this word also received a positive score and showed great importance for the classification, as in the explanation generated by LIME earlier. This may be because UOL is one of the largest online media companies in Brazil and is known for its reliable journalism, so citing this company can lend credibility to the news.
- ***Confira*** (check out): the word “Confira” is commonly used in news headlines; its positive score may indicate that this type of headline is more used in legitimate news.

1.2.1.4 Real news classified as fake

The real news incorrectly classified in Figure 6d did not present many highlights in the explanation generated by IG; the words showed very similar tones.

- ***Uma*** (a/one [feminine article]): this word had the greatest importance and a positive weight.
- Most words had a more negative score, with medium importance as can be noted by the lighter tone.

1.2.2 Model 2: FakeRecogna without stop words

1.2.2.1 Fake news classified as fake

Analyzing the fake news sample classified as fake by Model 2 in Figure 7a, it was possible to observe:

- ***alertapessoal*** (personal alert): this word showed greater importance and has the highest saturation, however it appeared in a purple tone, meaning it is between the false and real weight.

Figure 7 – Explanations generated by IG of Model 2

oi pessoal recebi alerta nao compramos peixe proximos dias local onde ficam
 armazenados ceagesp ficou inundado risco contrair doencas principalmente
 leptospirose lavar frutas verduras legumes bem antes descascar consumir nao comer
 nada cru salada vai alertapessoal bom dia alguns voces sabem trabalho distribuidora
 carnes congelados centro distribuicao fica ceagesp ontem impossivel dese trabalhar
 hoje vim aqui conferir toda situacao pessoal vdd puderem evitem frutas verduras
 proximos dias alem precos absurdos qualidade sera piores historia sera tudo
 descartado pura ilusao muita muita comida meio dessa lama lixo todoo
 realmente estragou certamente ira lixo demais gente certamente sera lavado
 colocado consumo porque sao toneladas toneladas alimentos prejuizo desses
 fornecedores serao imensos nesse pais onde dinheiro fala alto duvido tudo aqui sera
 jogado sujeira imensa bicho morto dar pau aqui entao fica alerta dica evitarem
 puderem

(a) Fake classified as fake

presidente jair bolsonaro fez discurso nesta tercafeira assembleia geral nacoes
 unidas onu fala apresentada meio video gravado causa pandemia covid reuniao onu
 neste ano baseada sede entidade nova york virtual bolsonaro disse brasil vitima
 campanha brutal desinformacao sobre amazonia pantanal segundo floresta
 amazonica umida so pega fogo bordas responsaveis queimadas sao indios caboclos
 disse tambem orientacoes pessoas fiquem casa pandemia quase levaram pais caos
 social bolsonaro disse floresta umida nao permite propagacao fogo interior
 incendios acontecem praticamente mesmos lugares entorno leste floresta onde
 caboclo indio queimam roçados busca sobrevivencia areas ja desmatadas

(b) Fake classified as real

exprefeito carioca eduardo paes dem entrevistado desta quinta-feira serie sabatinas
 promovidas folha spaulo uol candidatos prefeitura rio janeiro entrevista candidato
 criticou gestao atual prefeito marcelo crivella republicanos defendeu acusacoes
 corrupcao governo lupa verificou algumas declaracoes candidato confira governo
 crivella aumentos pedagio linha amarela dobro inflacaoeduardo paes candidato
 prefeitura rio janeiro sabatina

(c) Real classified as real

noticia bastante compartilhada redes sociais chegou portais regioes norte nordeste
 semana passada texto afirma candidata vereadora municipio presidente dutra
 maranhao localizado cerca km capital sao luis sido presa policia civil acusada
 oferecer sexo troca votos informacao fantasiosa segundo policia civil maranhao nao
 ha qualquer registro prisao candidata municipio nessas circunstancias noticia falsa
 afirma ainda policia encontrado celular candidata videos utilizaria garantia
 caracterizaria crime extorsao policia estima possam pessoas envolvidas nesse
 negocio

(d) Real classified as fake

- Most words did not show much importance for the classification, being at a very close saturation level.

1.2.2.2 Fake news classified as real

In Figure 7b, there is a fake news sample classified as real; it is possible to note:

- **video** (video): this word received a more positive score and great importance. The presence of the word **video** may indicate that the news includes visual evidence in the form of a video, hence receiving a positive score.
- **york** (York): reference to a city had a more positive score; the importance attributed to this word may be due to the fact that real news tends to be more complete, bringing information such as the location where events occurred.
- **brutal** (brutal): this word had a quite positive weight, however, it did not show much importance.

1.2.2.3 Real news classified as real

In Figure 7c there is a real news correctly classified; according to it, it is possible to note that:

- **exprefeito** (former mayor): this word showed the greatest importance; in this case, a reference to a political position had a positive score for the classification.
- **promovidas** (promoted): presented a more negative score; this may be associated with narratives involving false or misleading promotions or advances.
- **quinta-feira** (Thursday) and **janeiro** (January): these words did not show great importance but had positive weights, which may reinforce the idea that real news brings more information; thus, a temporal information tends to have a positive score.

1.2.2.4 Real news classified as fake

In Figure 7d there is a real news classified as fake; it is possible to note that:

- **compartilhada** (shared): this word was one of the most important, having a more negative score. In many cases, fake news or misinformation is widely shared on social networks or other online platforms. The term **compartilhada** may be associated with this type of dissemination of misleading information, leading the model to assign a negative score to the word.
- **envolvidas** (involved): this had a more balanced score between real and fake; however, it showed high importance.
- **votos** (votes): had a negative weight; this may be due to the fact that in many cases, fake news is related to electoral processes, voting, or politics.

1.2.3 Model 3: Fake.Br including stop words

1.2.3.1 Fake news classified as fake

Analyzing the samples classified by Model 3, starting with the fake news classified as fake in Figure 8a, it was possible to note:

- The words had very similar levels of importance.

Figure 8 – Explanations generated by IG of Model 3

Estagiário é flagrado "ao vivo" assistindo "filminho adulto" durante apresentação de telejornal. . Telespectadores que assistiam um noticiário da BBC transmitido na última segunda - feira (7) ficaram de queixo caído ! Eram quase 10h da noite quando a âncora Sophie Raworth falava sobre a vitória da equipe britânica de cricket sobre a África do Sul . Um detalhe inusitado conseguiu chamar mais atenção do que a própria notícia que estava sendo transmitida . Atrás da apresentadora havia um estagiário distraído brincando com seu computador [. . .] ele assistia um filminho de " sacanagem " . O resultado ? Tudo foi transmitido ao vivo ! Não poderia acabar de modo diferente . O estagiário foi afastado temporariamente de suas funções .

(a) Fake classified as fake

History Channel : Hitler forjou a própria morte e fugiu para a Argentina . Documentos comprovavam que Adolf Hitler e Eva Braun fugiram para a Argentina em um Submarino . O FBI revelou que o governo dos EUA sabia que Adolf Hitler estava vivo e , em 1984 , ele estava morando na Cordilheira dos Andes , muito tempo depois do fim da Segunda Guerra Mundial . (History Channel) Nos últimos 70 anos , o mundo foi levado a acreditar que Hitler teria morrido em 30 de abril de 1945 depois de se suicidar em seu bunker subterrâneo . De acordo com a história oficial , seu corpo foi descoberto e identificado pelos soviéticos e depois transportado para a Rússia . OS RUSSOS MENTIRAM ? Com a divulgação desses documentos do FBI , certamente parece que o líder mais notório da história mundial escapou da Alemanha para desfrutar uma vida pacífica no alto da Cordilheira dos Andes , América do Sul . Recentemente uma foto (abaixo) foi postada nas redes sociais sugerindo um " suposto " Hitler idoso , de 95 anos , posando com a namorada no Brasil em 1984 . Os recém - lançados arquivos do FBI afirmam que Adolf Hitler escapou através de um submarino com destino à Argentina , onde ele

(b) Fake classified as real

Traficante , primeira - dama e ex - guarda - costas em guerra de poder na Rocinha : a visão de autor de biografia de Nem . Para jornalista britânico Misha Glenny , autor de O dono do morro : um homem e a batalha pelo Rio , UPPs e ações militares , não têm sido capazes de reverter incapacidade de governos para lidar com guerra de facções . De um lado , o antigo guarda - costas e atual desafeto , que assumiu o poder do morro depois da prisão do mentor . De outro , a mulher do ex - chefe , condenada por tráfico , foragida , mas que , nas redes sociais , ostenta os cabelos pintados bem louros , o corpo bronzeado com biquínis e decotes , olhos escuros de grife e colares de ouro com pingente de coroa . Veja em tempo real a cobertura sobre a Rocinha Ainda é cedo para concluir que a invasão por traficantes no domingo e a disputa de poder na Rocinha , na zona sul do Rio de Janeiro , tenham sido ordenadas pelo traficante Antônio Francisco Bonfim Lopes , vulgo Nem , de dentro de uma prisão de segurança máxima , considera o jornalista britânico especializado em crime organizado Misha Glenny . Mas o racha interno e a crescente tensão entre a mulher e " herdeira " de Nem , Danúbia Rangel , e seu ex -

(c) Real classified as real

O ex - governador do Ceará , Cid Gomes (PDT) , afirmou em discurso ontem em Sobral , na sua festa de aniversário , que se ele estiver envolvido no esquema de corrupção da Petrobrás , o ministro do STF , Teori Zavascki , que relata a Lava Jato , é " como " ; o procurador - geral da República , Rodrigo Janot , é " ladrão " e o juiz Sérgio Moro , que atua nas investigações da primeira instância , é " picareta " . O nome de Cid Gomes aparece numa lista apreendida pela Lava Jato na sede da Odebrecht que esta sob investigação do STF . Cid aparece ao lado do valor de R \$ 200 mil e do codinome " falso " . Ouça o áudio :

(d) Real classified as fake

- **telejornal** (news broadcast): this word received a negative score, which may indicate that the model associated **telejornal** with sources of information that were discredited in the past for spreading misinformation.

1.2.3.2 Fake news classified as real

Analyzing the fake news classified as real in Figure 8b, it was possible to note:

- **History** (history): it is strange to note that this word had importance in two ways, once with a more positive weight and once with a more negative weight.
- **FBI** (FBI): again, this agency had a positive score; mentioning this organization adds more reliability to news.
- **forjou** (forged) and **Mentiram** (lied): two words associated with deceptive activities had importance in the prediction; however, they had positive scores, which may be due to the fact that in some situations these terms can indicate transparency and journalistic responsibility, showing that the story is exposing fraudulent or deceptive activities.

1.2.3.3 Real news classified as real

Analyzing Figure 8c, the real news classified as real showed:

- In this news, the words had very similar weights.
- **dama** (lady): the negative score attributed to this word may be explained by the association with the term “first lady” (which is the term used in the sample), which often appears in sensationalist news. This association suggests that the political figure in question may be the target of rumors or false narratives.
- **Misha Glenny, Danúbia Rangel, Antonio Francisco Lopes**: these names were associated with negative scores, possibly because they are frequently targets of fake news, or generally fake news targets people, so the use of proper names receives this score.

1.2.3.4 Real news classified as fake

Analyzing Figure 8d, the real news classified as fake showed:

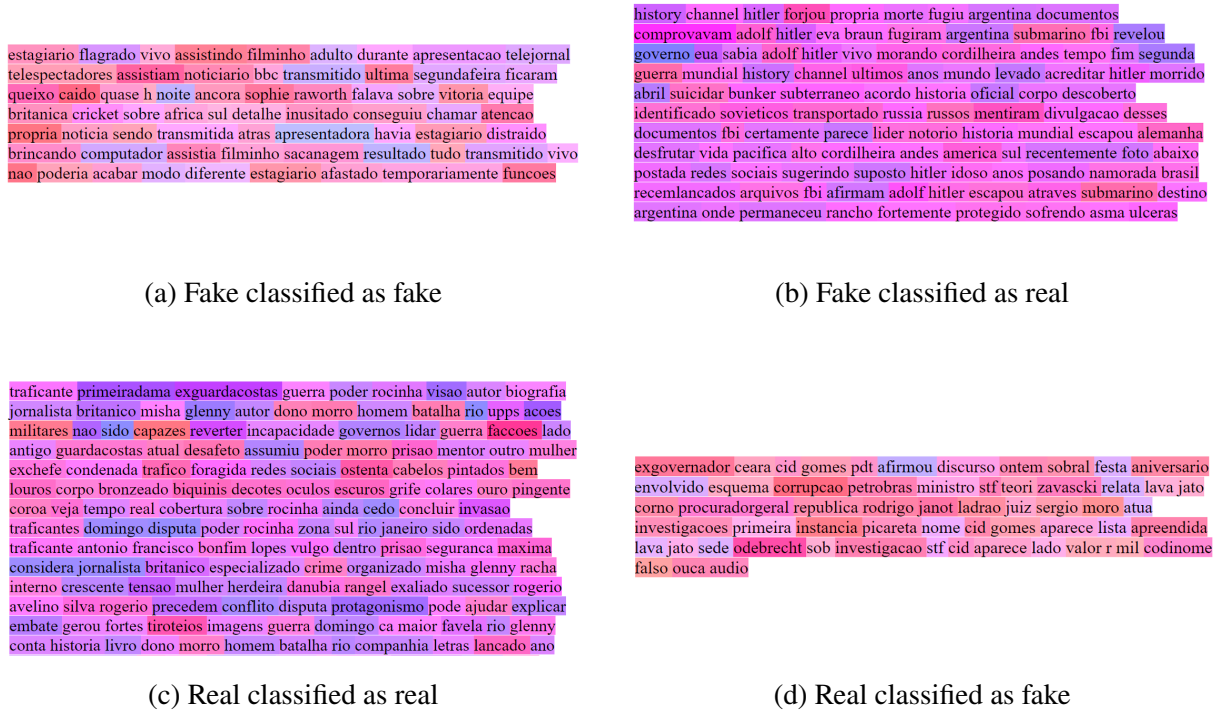
- **Sobral** (Sobral): this word had a negative score; it refers to a city by this name in the news, but Sobral is also a common surname in Brazil, so the model may have associated a proper name with fake news.
- **investigação** (investigation): the mention of this word in news may suggest there is an active search for truth and a responsible approach to uncover accurate information. This can increase readers’ trust in the news’ veracity, so the word receives a positive score.
- **ladrão** (thief): a word with negative connotation, it had a negative score, which may indicate that this word is often used to unfairly accuse people in fake news. However, crimes are commonly reported, and the word **ladrão** should not have such a negative score.

1.2.4 Model 4: Fake.Br without stop words

1.2.4.1 Fake news classified as fake

Analyzing now the samples classified by Model 4, starting with the fake news classified as fake in Figure 9a, it was possible to note:

Figure 9 – Explanations generated by IG of Model 4



- **atenção** (attention): this word had a negative score; if the word **atenção** is used in a sensationalist or alarming context, it may indicate that the news is fake. News that tries to grab the reader's attention in an exaggerated or misleading way can be seen as less trustworthy.
- **não** (no/not): this word received a negative score, it is a negation word; however, its highlight with a negative weight may not mean much.

1.2.4.2 Fake news classified as real

Analyzing a fake news classified as real in Figure 9b, it was possible to observe:

- **forjou** (forged), **mentiram** (lied): in this model, these words also received a negative score.
- **governo** (government): this word received a positive score, which may be because the term **governo** often refers to government institutions responsible for providing official and reliable political information.

1.2.4.3 Real news classified as real

Analyzing a real news classified as real in Figure 9c, it was possible to observe:

- ***primeiradama*** (first lady): in this model without *stop words*, it is interesting to note that the word ***primeiradama*** had a positive weight.
- ***exguardacostas*** (former coast guard): this word received a positive score; the use of detailed and specific descriptions may contribute to a more accurate and informative analysis.
- ***domingo*** (Sunday): again, a temporal reference word received a positive score.
- ***glenny*** (Glenny): this time only a surname received a negative score.

1.2.4.4 Real news classified as fake

In Figure 9d is the real news misclassified; from it, it was possible to observe:

- ***exgovernador*** (ex-governor): reference to a political figure received a negative score.
- ***corrupção*** (corruption): this word had great importance and showed a negative weight; this may be because many fake news accuse politicians of corruption due to political bias.
- ***odebrecht*** (Odebrecht): this word received a positive score and a high level of importance; this may be because Odebrecht is a company involved in several corruption scandals in various countries, which can generate distrust regarding news mentioning the company's name.